

SURAJ GUPTA

AI Engineer & Full-Stack Developer

+91 6263117774 | surajg7774@gmail.com | github.com/surajg7774 | linkedin.com/in/suraj-gupta-493502210 | Khargone, Madhya Pradesh, India

SUMMARY

Building production-ready AI applications and scalable full-stack software that solve real-world problems through modern engineering. Comfortable across the stack: enterprise-grade backend systems (Spring Boot, Spring Security, REST APIs, role-based access control), React front ends, and applied AI pipelines (RAG, OCR, vector databases, NLP). PG-DAC graduate (CDAC ACTS, 2025), currently in CDAC ACTS' PGCP-AI program. Seeking full-time opportunities as an AI Engineer or Full-Stack Developer.

CORE SKILLS

Languages: Java, Python, JavaScript, SQL

Backend / Frontend: Spring Boot, Node.js, FastAPI | React, TypeScript, Material UI, HTML, CSS

AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision, Transformers, RAG, LangChain, LangGraph

Databases: MySQL, ChromaDB, MongoDB

Cloud / Dev Tools: Vercel, Railway, AWS | Git, Docker, Postman

PROJECTS

SG Krashi — Smart Agriculture Ecosystem Platform | *Solo Project* | 2 months

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Tech: React 19, TypeScript, Vite, MUI, Java 21, Spring Boot, Spring Security, JWT, MySQL, Razorpay, Google Gemini, Cloudinary, Brevo

- Architected and built a live, multi-vertical platform (organic farming, dairy, equipment rental, farm stays, product store, crop marketplace) solo across 20 formally-scoped modules, with real Razorpay payments in production.
- Built AI Crop Doctor: photo-based crop disease diagnosis via Google Gemini with a retrieval-augmented (RAG) knowledge layer, confidence bands instead of fabricated percentages, and a declared-crop mismatch cross-check.
- Designed a generic booking engine powering both Equipment Rental and Farm Stay, plus a recommendation system and predictive analytics (revenue forecasting, stock-out risk).
- Impact: A real, live platform running six business lines for an actual agriculture business — not a tutorial project.

Equipment Diagnosis System (ServiceDiagnose AI) | *Solo Project* | 1 month

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Tech: Python, FastAPI, ChromaDB, SQLAlchemy, Pydantic, Groq LLM, ONNX embeddings, React, Vite, Render, Vercel

- Built a RAG-based fault-diagnosis assistant for field technicians: retrieves similar past cases from ChromaDB and has an LLM (Groq in production, Ollama locally) write a tailored diagnosis, action and severity in English, Hindi or Hinglish.
- Added iterative diagnosis (up to 4 fixes, then human escalation), text + photo diagnosis via a vision model, and separate retrieval vs LLM confidence.
- Designed a human-in-the-loop learning loop (feedback → provisional fix → technician-verified; failed fixes steer answers) over 183 records, 19 categories.

Cross-Border Ingredient Safety & Compliance System | *Team Project* | 1 month

Tech: Python, FastAPI, Streamlit, LangGraph, ChromaDB, Sentence-Transformers, PaddleOCR, OpenCV, Pydantic, Pytest

- Checks food ingredients against different markets' regulations: label images are read with PaddleOCR + OpenCV (or ingredients typed in), then normalized to standard INS / E-number identities.
- Built a RAG pipeline: regulatory sources (Codex GSFA, EU, UK, Canada, US, openFDA recalls) embedded with Sentence-Transformers into ChromaDB; retrieved evidence is fed to the LLM instead of its internal knowledge.
- LangGraph runs market agents (EU, UK, Canada, US) and a recall agent; a deterministic Rule Engine, not the LLM, decides Safe / Restricted / Prohibited.
- A Risk Aggregator merges market checks into one report with human review for uncertain cases; Streamlit dashboard, PDF/CSV/JSON export.

Bharat Fix (Complaint Management System) | *Team Project* | 1 month

Tech: Java, Spring Boot, React, MySQL, Spring Security, JWT, Google Maps API

- Built a complaint management system with Google Maps location pinning for citizen-reported issues.
- Implemented OTP-based authentication and role-based access control (User/Staff/Admin) with Spring Security and JWT.
- Developed RESTful APIs for status tracking, complaint assignment and workload management.
- Impact: A working model for transparent civic complaint resolution instead of issues disappearing into a queue.

EDUCATION

PGCP-AI (Post Graduate Certification Program in Artificial Intelligence) — CDAC ACTS	Feb 2026 batch — Ongoing
PG-DAC (Post Graduate Diploma in Advanced Computing) — CDAC ACTS, Pune	2025 · 76.4%
B.Tech, Computer Science & Engineering — Sushila Devi Bansal College of Engg, Indore (RGPV Bhopal)	2024 · 77.6%
Class XII (PCM) — Govt. Devi Ahilya Excellence H.S. School No.1, Khargone (M.P. Board)	2020 · 66.4%
Class X — Govt. Devi Ahilya Excellence H.S. School No.1, Khargone (M.P. Board)	2018 · 74.6%

CERTIFICATIONS

PGCP-AI (Post Graduate Certification Program in Artificial Intelligence) — CDAC ACTS

EXTRACURRICULAR

- Active member, E-Cell (Entrepreneurship Cell) — organized workshops, business plan competitions, and speaker sessions.
- Sports Captain, School | Volunteer, National Service Scheme (NSS).